

Module 01: Systems — Mathematical Modeling of Systems

Learning Objectives

1. Represent a real-world system as a set of **state variables**, **parameters**, and **update rules**.
2. Distinguish between **open systems** (exchange energy/information with the environment) and **closed systems**.
3. Connect the concept of a system's **boundary** to the mathematical idea of a Markov blanket.

Introduction

In the Everyday Life and Biology & Health units, you learned to *recognize* systems everywhere — in your school, your body, and your social life. Now we put numbers on things. A mathematical model is a translation of a real system into variables and equations. This is the foundation of all science, engineering, and data science.

Consider a simple population model: $P_{t+1} = r \cdot P_t \cdot (1 - P_t / K)$. Here, P_t is the current population (a **state variable**), r is the growth rate (a **parameter**), and K is the carrying capacity (a **constraint** — it acts like a boundary on how big the population can get). This single equation captures the prediction-correction loop of Active Inference: the population "predicts" exponential growth, but the carrying capacity provides a corrective signal, pulling the population back toward equilibrium.

Key Concepts

1. State Variables and State Space

A **state variable** is any quantity that describes the current condition of a system. Temperature, position, population size, bank balance — all are state variables. The set of all possible values forms the **state space**. A system's trajectory through state space is like a path through a map.

2. Parameters and Constraints

Parameters are values that shape the system's behavior but do not change during the model's run (like the growth rate r). **Constraints** are boundaries on what the system can do. In Active Inference, the Markov blanket *is* the constraint: it determines which external information the system can access.

3. Equilibria and Attractors

An **equilibrium** is a state where the system stays put. A **stable equilibrium** (an attractor) means the system returns there after a perturbation — like body temperature returning to 98.6°F . Attractors are the mathematical expression of the idea that living systems resist surprise.

Applications

- **Algebra:** A system of linear equations ($2x + 3y = 12$, $x - y = 1$) constrains the values of x and y . The solution set is the system's equilibrium. Adding a third equation (new evidence!) further narrows the possibilities — just like sensory data reduces an agent's uncertainty.
- **Population Ecology:** The logistic equation above models a self-regulating population. Students can simulate it in a spreadsheet and watch how different values of r produce stability, oscillation, or chaos.

Discussion Questions

1. What are the state variables in your school's cafeteria system? What are the constraints?

2. Can a system have more than one equilibrium? Give an example.

Summary

Mathematical modeling translates real systems into state variables, parameters, and rules. Equilibria and attractors describe how systems resist surprise. This mathematical toolkit underlies all of Active Inference.

References

- Strogatz, S. (2015). *Nonlinear Dynamics and Chaos*. Chapter 1.